**DP014**
*Physics 1*
*Semester I*
*Session 2024/2025*
*2 hour*

**DP014**
**Fizik 1**
**Semester I**
**Sesi 2024/2025**
**2 jam**

**K**EMENTERIAN
**P**ENDIDIKAN
**M**ALAYSIA

# KOLEJ MATRIKULASI PERAK

## PRA-PSPM
***SET 1***

**JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.**
*DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.*

Name: ____________________

Matric. No.: ____________________

Class: __________

| NO. | MARKS | |
|---|---|---|
| 1 | 8 | |
| 2 | 12 | |
| 3 | 9 | |
| 4 | 12 | |
| 5 | 9 | |
| 6 | 12 | |
| 7 | 18 | |
| TOTAL | **80** | |

**INSTRUCTION TO CANDIDATE:**

This question paper consists of **7** questions. Answer **all** the questions.
The use of electronic calculator is permitted.

## LIST OF SELECTED CONSTANT VALUES

| | | |
|---|---|---|
| Speed of light in vacuum | $c$ | $= 3.00 \times 10^{8}$ m s$^{-1}$ |
| Permeability of free space | $\mu_o$ | $= 4\pi \times 10^{-7}$ H m$^{-1}$ |
| Permittivity of free space | $\varepsilon_o$ | $= 8.85 \times 10^{-12}$ F m$^{-1}$ |
| Elementary charge magnitude | $e$ | $= 1.60 \times 10^{-19}$ C |
| Planck constant | $h$ | $= 6.63 \times 10^{-34}$ J s |
| Electron mass | $m_e$ | $= 9.11 \times 10^{-31}$ kg |
| | | $= 5.49 \times 10^{-4}$ u |
| Neutron mass | $m_n$ | $= 1.674 \times 10^{-27}$ kg |
| | | $= 1.008665$ u |
| Proton mass | $m_p$ | $= 1.672 \times 10^{-27}$ kg |
| | | $= 1.007277$ u |
| Deuteron mass | $m_d$ | $= 3.34 \times 10^{-27}$ kg |
| | | $= 2.014102$ u |
| Molar gas constant | $R$ | $= 8.31$ J K$^{-1}$ mol$^{-1}$ |
| Rydberg constant | $R_H$ | $= 1.097 \times 10^{7}$ m$^{-1}$ |
| Avogadro constant | $N_A$ | $= 6.02 \times 10^{23}$ mol$^{-1}$ |
| Boltzmann constant | $k$ | $= 1.38 \times 10^{-23}$ J K$^{-1}$ |
| Gravitational constant | $G$ | $= 6.67 \times 10^{-11}$ N m$^{2}$ kg$^{-2}$ |
| Free-fall acceleration | $g$ | $= 9.81$ m s$^{-2}$ |
| Atomic mass unit | 1 u | $= 1.66 \times 10^{-27}$ kg |
| | | $= 931.5 \ \frac{\text{M}e\text{V}}{c^2}$ |
| Electron volt | 1 $e$V | $= 1.6 \times 10^{-19}$ J |
| Constant of proportionality for Coulomb's law | $k = \frac{1}{4\pi\varepsilon_o}$ | $= 9.0 \times 10^{9}$ N m$^{2}$ C$^{-2}$ |
| Atmospheric pressure | 1 atm | $= 1.013 \times 10^{5}$ Pa |
| Density of water | $\rho_w$ | $= 1000$ kg m$^{-3}$ |

**LIST OF SELECTED FORMULAE**
***SENARAI RUMUS TERPILIH***

1. $v = u + at$
2. $s = ut + \frac{1}{2}at^2$
3. $v^2 = u^2 + 2as$
4. $s = \frac{1}{2}(u + v)t$
5. $v = u - gt$
6. $v^2 = u^2 - 2gs$
7. $s = ut - \frac{1}{2}gt^2$
8. $p = mv$
9. $J = F\Delta t$
10. $\vec{J} = \Delta\vec{p} = m\vec{v} - m\vec{u}$
11. $f_s \leq \mu_s N$
12. $f_k = \mu_k N$
13. $W = \vec{F} \cdot \vec{s} = Fs\cos\theta$
14. $K = \frac{1}{2}mv^2$
15. $U = mgh$
16. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$
17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$
18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$
19. $s = r\theta$
20. $v = r\omega$
21. $a_t = r\alpha$
22. $\omega = \omega_o + \alpha t$
23. $\theta = \omega_o t + \frac{1}{2}\alpha t^2$
24. $\omega^2 = {\omega_o}^2 + 2\alpha\theta$
25. $\theta = \frac{1}{2}(\omega_o + \omega)t$
26. $\tau = rF\sin\theta$
27. $\frac{Q}{t} = -kA\left(\frac{\Delta T}{L}\right)$
28. $v_{rms} = \sqrt{< v^2 >}$
29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$
30. $\Delta U = Q - W$

1 (a) An object has mass of 571.5 g and volume of 2250 $cm^3$. Calculate the density of the object.

[3 marks]

(b)

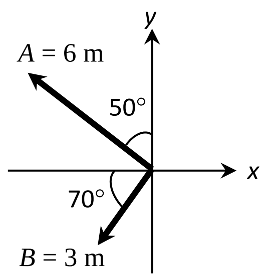


**FIGURE 1**

**FIGURE 1** shows two displacement vectors A and B. Calculate the magnitude and direction of the resultant vector, **R** = **A** + **B**.

[5 marks]

2 (a)

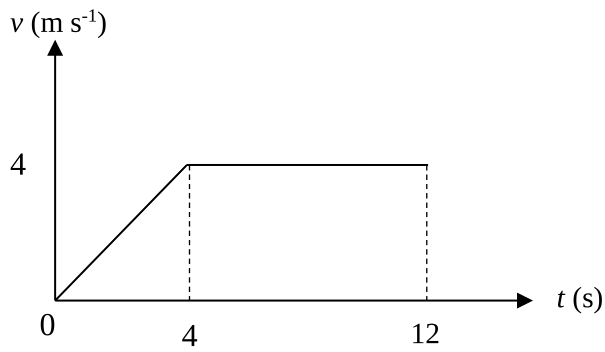


**FIGURE 2**

**FIGURE 2** shows the velocity-time graph describing the motion of an object. Sketch the

(i) acceleration-time graph

(ii) displacement-time graph

[8 marks]

(b) A stone is thrown vertically upward with a speed of 5 m $s^{-1}$. After 1.2 s, calculate the

(i) displacement

(ii) velocity of the stone.

[4 marks]

3 (a) Two balls with masses of 0.4 kg and 0.6 kg respectively are thrown in such a manner that they collide head-on. They stick together and move with common velocity *v*. The initial velocity for each ball is 15 m $s^{-1}$. Calculate the velocity *v*.

[3 marks]

(b) An object A of mass 2 kg moves to the right at a speed of 5 m $s^{-1}$. Object B of mass 4 kg moves to the left at a speed of 8 m $s^{-1}$. Calculate the velocity of A if object B moves at velocity of 4 m $s^{-1}$ to the right after the collision.

[3 marks]

(c) A net force of 8.0 N acts on an 18 kg body for one minute.

(i) Calculate the impulse due to the force.

(ii) If the final velocity is 60 m $s^{-1}$, calculate the initial velocity of the body.

[3 marks]

4 (a)

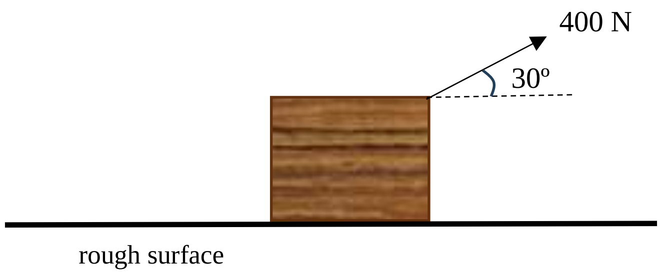


**FIGURE 3**

**FIGURE 3** shows a 70 kg box is pulled by a 400 N force at an angle of 30º to the horizontal. The coefficient of kinetic friction is 0.50.

(i) Sketch with label the free body diagrams which show all the forces that act on the box.

(ii) Calculate the normal force.

(iii) Calculate the frictional force.

(iv) Calculate the acceleration of the box.

[9 marks]

(b) A horizontal force of 140 N is needed to pull a 60 kg box across the horizontal floor at constant speed. Calculate the coefficient of kinetic friction between floor and box.

[3 marks]

5 (a) A stone at the end of a string of length 50 cm is whirled in a horizontal circle on a smooth surface. The speed of the stone is 0.3 m $s^{-1}$.

(i) Calculate the angular velocity of the stone.

(ii) How many revolutions does the stone make in 3 minutes?

[4 marks]

(b) A 0.2 kg ball, attached to the end of a string, is rotated in a horizontal circle of radius 1.5 m on a frictionless table surface. If the tension of the string is 50 N,

(i) draw the free body diagram to show all the forces acting to the ball.

(ii) calculate the speed of the ball.

[5 marks]

6 (a) A car has wheel of diameter 50 cm. It starts from rest and accelerates uniformly to a speed of 16 m $s^{-1}$ after 8 s. Calculate the

(i) tangential acceleration,

(ii) angular acceleration,

(iii) angular displacement, and

(iv) number of rotations one wheel makes in this time.

[8 marks]

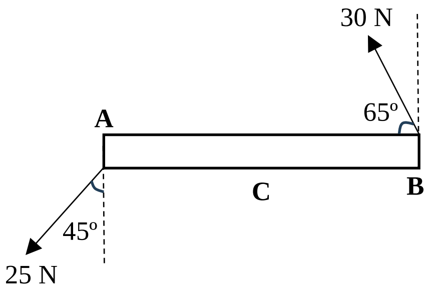


**FIGURE 4**

(b) Calculate the net torque on the 4 m uniform beam of mass 5 kg in **FIGURE 4** by taking moment about the left end of the beam, **A**.

[4 marks]

7 (a) Two rooms, each a cube 4 m on a side, share a 12 cm thick insulated brick wall. Since there are several 112 W light bulbs in one room, the air is at 30 ºC, while in the other room is at 10 ºC.

(i) calculate the number of 112 W light bulbs are needed to maintain the temperature difference across the wall.

(ii) sketch the graph of temperature against distance of the wall.

(Given coefficient of thermal conductivity of brick is 0.84 W $m^{-1}$ $°C^{-1}$)

[6 marks]

(b) A nitrogen gas is at 0 ºC. Calculate the

(i) mass of one nitrogen molecule.

(ii) root mean square speed a nitrogen molecule.

(iii) new temperature that it must be raised to double the root mean square speed of its molecules.

(Given that the molar mass of nitrogen molecule is 28 g $mol^{-1}$)

[6 marks]

(c) An ideal gas undergoes two thermodynamics processes as follows.

Process PQ: The gas does 5 J of work while expanding adiabatically.

Process QR: Then, 8 J of work in done on the gas while compressing isothermally to the initial volume.

(i) Calculate the change in internal energy during process PQ.

(ii) Calculate the heat during the process QR.

(iii) Sketch a p-V diagram to describe the thermodynamic processes PQR.

[6 marks]

## END OF QUESTION PAPER